MODULE 5 · LECTURE 25

CS 6530 Applied Cryptography

IKEv2: Establishing and Diagnosing IPsec SAs

Module 5 – IPsec, IKE and Secure Gateways

Where Are the IPsec Peers?

Deployment	Possible IPsec Peers	Traffic Being Protected
Enterprise site-to-site VPN	Branch security gateway ↔ headquarters security gateway	Selected traffic between the two office networks
Operator network access	Remote network endpoint ↔ operator security gateway	Selected traffic toward the network behind the gateway
O-RAN E2 interface	Near-RT RIC ↔ E2 node, such as an O-DU, O-CU-CP, or O-CU-UP	Selected IP traffic carrying E2 signalling
O-RAN D2 interface	O-DU ↔ O-DU, when IPsec is deployed	Selected IP traffic exchanged across D2

O-RAN = Open Radio Access Network

IPsec peers apply and remove IPsec protection.
They may be security gateways, or the communicating network functions may perform IPsec themselves.

SPD decision: PROTECT

↓

Suitable Security Associations and keys are required

How do the two IPsec peers establish the security state that ESP needs?

Simplified deployment patterns, at a glance



What Must the Two Peers Establish?

Selected traffic requires protection
SPD decision: PROTECT

Selected traffic protected by ESP

PEER A

JOB 1 — TRUSTED IKE RELATIONSHIP

- Agree on cryptographic parameters for IKE.
- Exchange Diffie–Hellman key material and nonces.
- Authenticate each other using an agreed method.

Result: An authenticated IKE SA provides the keys and security state used to protect subsequent IKE exchanges.

JOB 2 — PROTECTION FOR SELECTED TRAFFIC

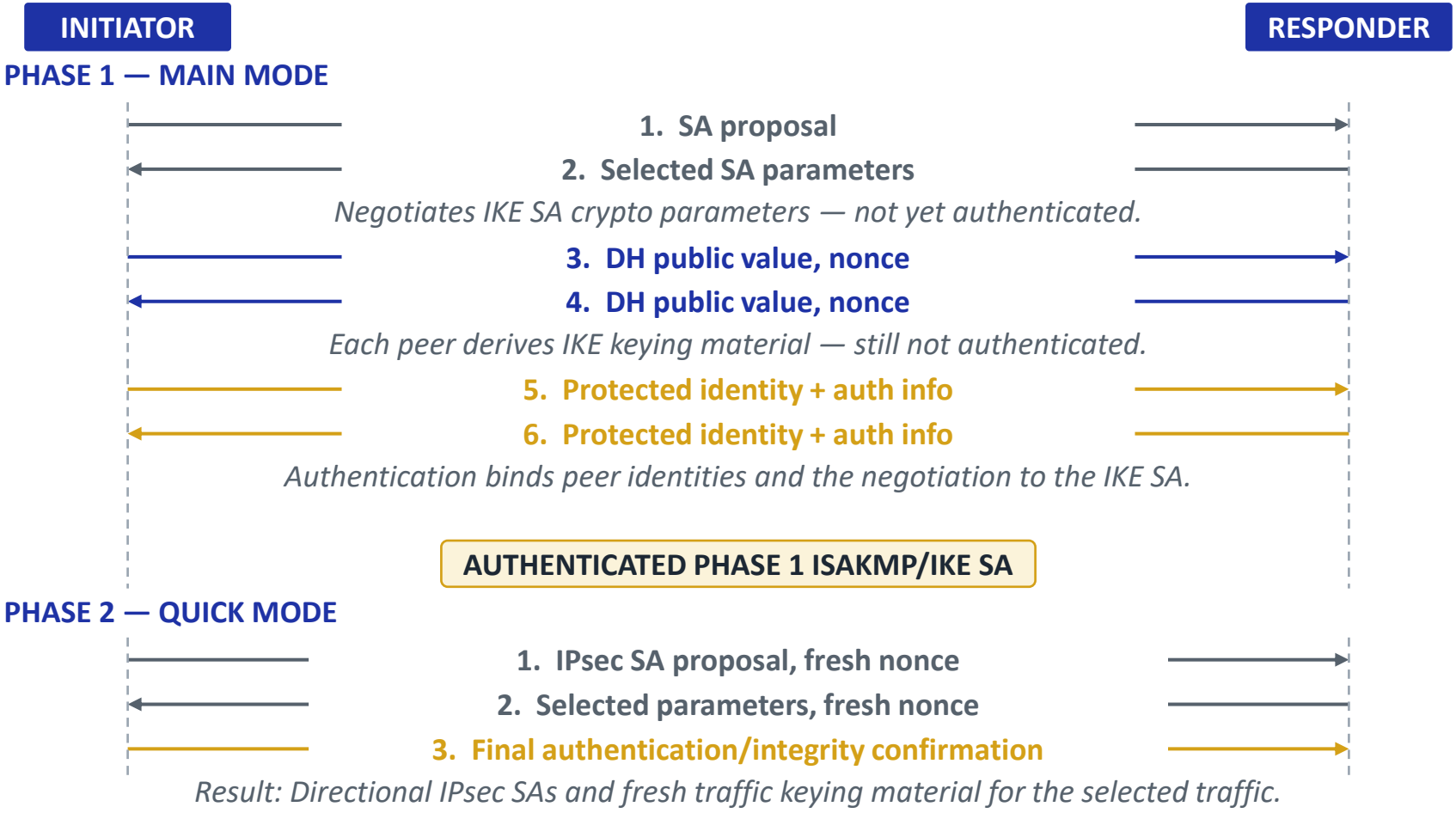
- Agree on which traffic is to be protected.
- Agree on IPsec/ESP parameters.
- Obtain traffic keys and establish directional IPsec SAs.

Result: IPsec SAs provide the state that ESP uses to protect packets. In IKEv2, the IPsec SAs established together belong to a CHILD SA.

PEER B

Policy selects traffic → established SAs supply security state → ESP protects packets

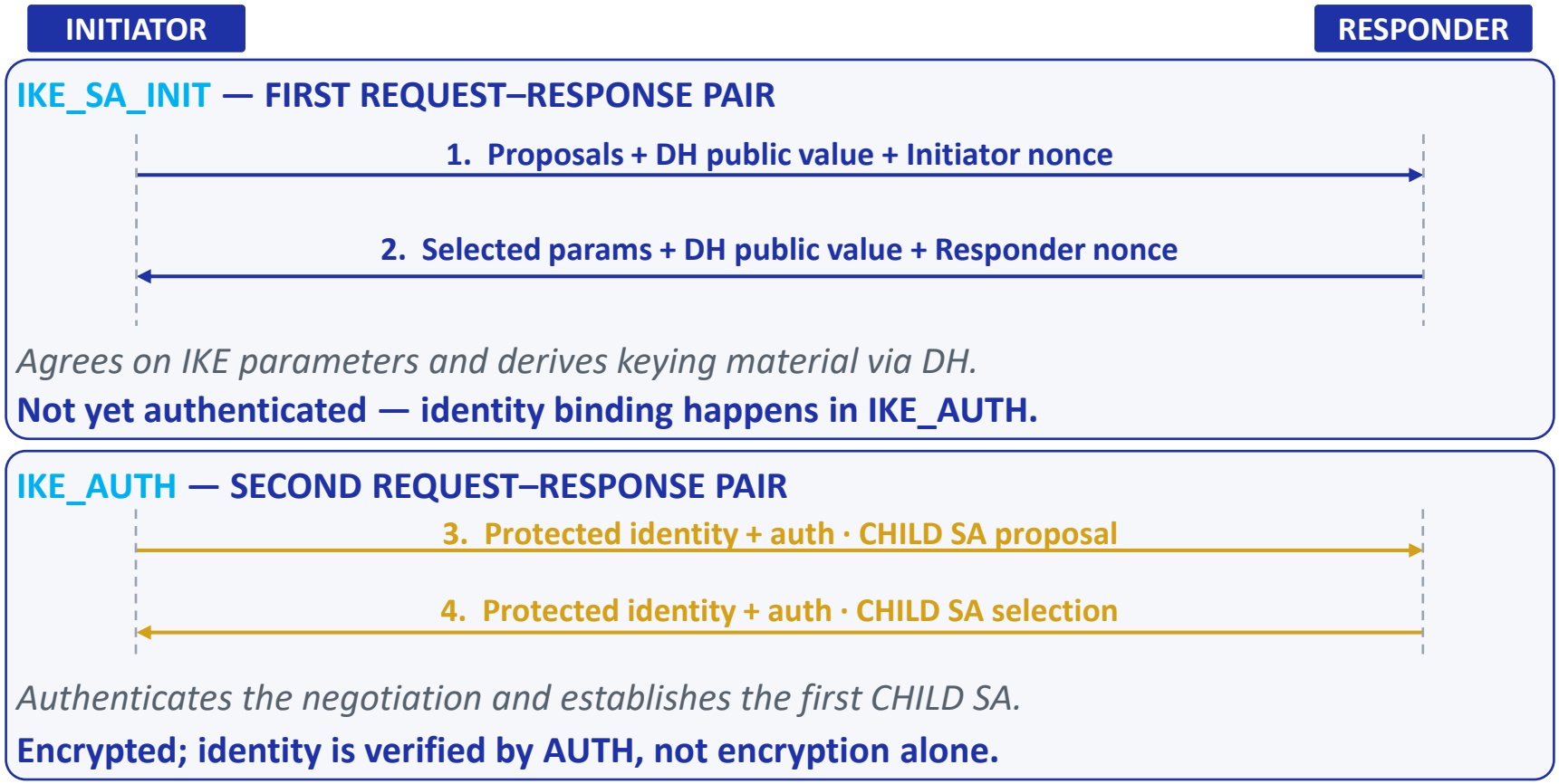
How Did IKEv1 Establish IPsec SAs?



Protected by the established Phase 1 SA; an additional DH exchange is optional in Quick Mode.

One Phase 1 SA can protect multiple Quick Mode negotiations.

How Does IKEv2 Organize the Initial Exchange?



After initial establishment

- CREATE_CHILD_SA — create or rekey a CHILD SA, or rekey the IKE SA.
- INFORMATIONAL — control messages, such as deletion or liveness checks.

Four messages describe the normal initial exchange. Retries, cookies, EAP authentication, and some other cases can add messages.

What Does IKE_SA_INIT Establish?

INITIATOR

Initiator → Responder

`SAi1 | KEi | Ni`

SAi1: proposed IKE crypto suite

KEi: initiator's DH public value

Ni: initiator's nonce

RESPONDER

Responder → Initiator

`SAr1 | KEr | Nr`

SAr1: selected IKE crypto suite

KEr: responder's DH public value

Nr: responder's nonce

Both sides derive the same DH secret g^ir — the public values and nonces cross the network; the secret itself is never transmitted.

`SKEYSEED = prf(Ni | Nr, g^ir)`

`SK_d | SK_ai | SK_ar | SK_ei | SK_er | SK_pi | SK_pr = prf+(SKEYSEED, Ni | Nr | SPIi | SPIr)`

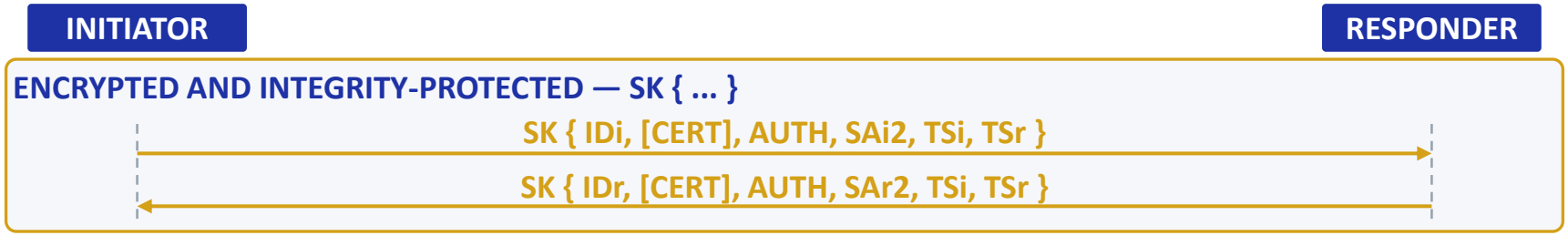
PRF = Pseudorandom Function · prf+ = iterative PRF-based key expansion · | = concatenation · SPIi/SPIr = the IKE SA SPIs

Keys for protecting IKE_AUTH now exist. Peer authentication has not yet completed.

Derived secret	Purpose
SK_d	CHILD SA keying material
SK_ai, SK_ar	Per-direction IKE integrity (if used)
SK_ei, SK_er	Per-direction IKE confidentiality
SK_pi, SK_pr	Used in each peer's AUTH calculation

i/r = per-direction (no single key protects both) · an AEAD transform folds integrity into encryption — no separate integrity transform.

How Does IKE_AUTH Authenticate the Peers?



Identity and proof

- IDi / IDr: state the initiator's / responder's identity.
- AUTH: proves identity and authenticates the earlier negotiation.
- [CERT]: optional certificate for certificate-based authentication.

First CHILD SA negotiation

- SAi2 / SAr2: proposed / selected IPsec-ESP parameters.
- TSi / TSr: traffic selectors negotiated for the CHILD SA (derived from / checked against local IPsec policy).
- Normally establishes the first CHILD SA within IKE_AUTH.

Method	What the peer proves
PSK	Knowledge of the configured shared secret
Certificate	Possession of the private key for the certificate

A certificate/encryption alone isn't proof — AUTH must verify.

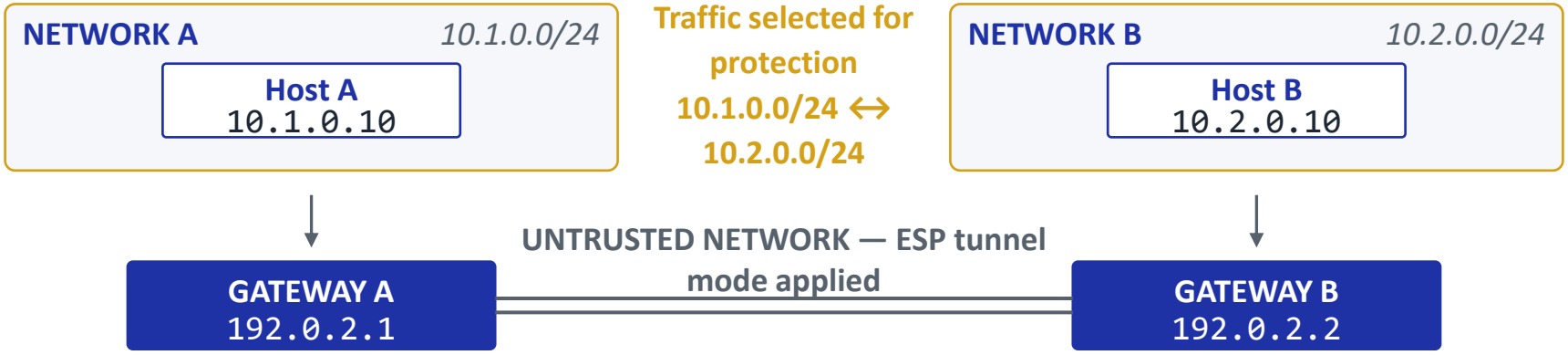
The DH values were visible during IKE_SA_INIT. How does IKE_AUTH stop silent replacement?

AUTH binds back to IKE_SA_INIT and the peer identity — a substitution breaks the proof.

Authenticated IKE SA — protects IKE exchanges

First CHILD SA — supplies the ESP SAs

Which Peers and Traffic Are We Protecting?



IKEv2 peers: Gateway A ↔ Gateway B — negotiate and maintain the IKE SA and CHILD SA.

Original packet

```
10.1.0.10  →  10.2.0.10
```

On the untrusted network

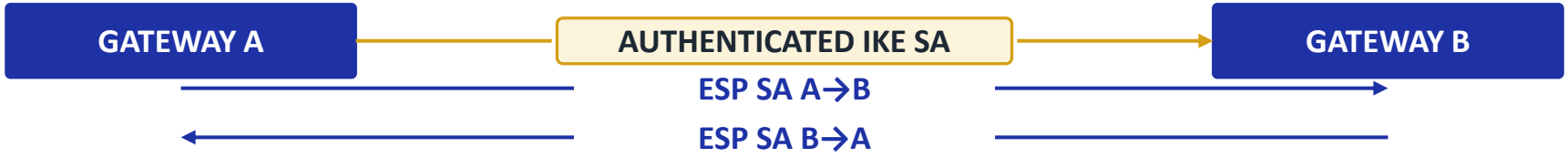
```
192.0.2.1  →  192.0.2.2  (outer)
ESP { 10.1.0.10 → 10.2.0.10 }
```

This illustrative deployment is used in the following SA and diagnosis diagrams.

Host addresses identify the communicating systems; gateway addresses identify the IKE/IPsec peers and the outer tunnel endpoints — the host addresses are not IKE peer addresses.

IKEV2 · FIRST CHILD SA

What Does IKE_AUTH Establish for ESP?



Each SA has its own SPI, keys, sequence-number counter, and inbound replay state (L24's unidirectional SA model).

Accepted traffic selectors

```
TSi:  10.1.0.0/24
TSr:  10.2.0.0/24
```

TSi/TSr name this exchange's initiator/responder, not Network A/B permanently.

Accepted ESP parameters

- ESP transform / algorithms
- Tunnel mode (this deployment)
- Lifetime and related SA parameters
- Directional keying material and SPIs

```
KEYMAT = prf+(SK_d, Ni | Nr)
```

SK_d is from the Slide-5 IKE key derivation; Ni/Nr are the IKE_SA_INIT nonces. KEYMAT supplies separate keying material per traffic direction.

IKE SA — protects IKE negotiation and management messages

CHILD SA — the IPsec SAs that protect selected traffic with ESP

The IKE SA's SPIs and the ESP SAs' SPIs are different associations — application packets flow through the ESP tunnel, never through the IKE SA itself.

What Can We Verify in the IKE Captures?

Capture peer A: 2003:51:6012::4 · Capture peer B: 2003:51:6012::2 · Transport: UDP port 500

LEFT — IKEv1 CAPTURE

Frames	Exchange	Observation
2–7	Main Mode	Six messages: Phase 1 relationship
8, 10, 12	One Quick Mode	Three messages under Phase 1

Frames 9, 11, 13 are a separate Quick Mode exchange, not part of this selected path.

RIGHT — IKEv2 CAPTURE

Frames	Exchange (MID)	Observation
2–3	IKE_SA_INIT (0)	Proposal, DH exchange, nonces
4–5	IKE_AUTH (1)	Protected IKE_AUTH messages
6–7	CREATE_CHILD_SA (2)	Later SA-management exchange

CREATE_CHILD_SA is a later exchange, not part of the initial four-message sequence.

Visible here: peer addresses, exchange types, Message IDs, SPIs, and the shift to protected payloads — not decrypted contents, PSK vs. certificate, or add-vs-rekey for CREATE_CHILD_SA.

Small final observation: ESP packets follow the IKE negotiation in both captures.

Slides 7–8: illustrative IPv4 site-to-site deployment. Slide 9: supplied IPv6 packet recordings.

How Are SAs Added or Rekeyed?

ESTABLISHED IKE SA

CHILD SA -- carries ESP traffic

`CREATE_CHILD_SA request → response → one of three outcomes:`

Purpose of the exchange	What is established
Add another protected flow	New CHILD SA pair, own selectors
Rekey an existing CHILD SA	Replacement CHILD SA: new keys & SPIs
Rekey the IKE SA	Replacement IKE SA: new keys & SPIs

Old SA carries traffic → **Replacement SA negotiated** → **Traffic moves to replacement** → **Old SA deleted**

Old and new SAs may briefly coexist during the transition.

- CREATE_CHILD_SA: a protected exchange either peer can initiate.
- New/rekeyed CHILD SAs get fresh nonces; DH is optional.
- IKE SA rekey needs a new DH exchange, not reauth -- reauth needs a new initial exchange.

Keep “add CHILD SA,” “rekey CHILD SA,” and “rekey IKE SA” visibly separate -- a CREATE_CHILD_SA packet's exchange name alone does not say which one occurred.

IKEV2 · NAT TRAVERSAL

What If a NAT Is Between the Peers?



NAT changes the outer address/port; the peers still maintain IKE and ESP through that mapping.

DETECT: IKE_SA_INIT notifications flag address/port translation along the path.

TRAVERSE: subsequent IKE traffic and UDP-encapsulated ESP shift to UDP port 4500.

MAINTAIN: a NAT keepalive preserves an otherwise idle UDP mapping.

Without UDP encapsulation	With NAT traversal
Outer IP → ESP	Outer IP → UDP 4500 → ESP
IKE commonly uses UDP 500	IKE on 4500 has a 4-byte Non-ESP Marker separating it from ESP.

UDP 4500 is a useful clue but does not alone prove a NAT exists -- the Slide 9 captures do not show this NAT traversal sequence.

Common for a remote client behind a home or enterprise NAT; the O-RAN E2/D2 examples do not inherently imply one.

IKEV2 · FAILURE DIAGNOSIS

Where Did Establishment Fail?

IKE_SA_INIT response → **Peer authentication** → **CHILD SA installed** → **Protected traffic**

Last observation	First things to examine
No IKE_SA_INIT response	Address/route reachability, UDP 500/4500 filtering; request reaching responder?
IKE_SA_INIT errors/stalls	Proposal compatibility, chosen algorithms/DH group; check any Notify.
IKE_AUTH fails to authenticate	Identities, PSK/cert config, trust/validity; check endpoint error/logs.
IKE SA up, no usable CHILD SA	ESP proposals, selector overlap, policy; check selectors and SA state.
CHILD SA up, traffic fails	Packet vs. SPD/route; ESP SAs/SPIs both ways, filtering, NAT mapping.

Diagnose from the last confirmed checkpoint, then compare evidence at both peers. A packet capture shows exchanges and outer flow; SA state and logs at the endpoints cover encrypted details and policy.

Traffic between 10.1.0.0/24 and 10.2.0.0/24 must match negotiated selectors and policy before protection applies. Do not blame a specific PSK, certificate, selector, or firewall fault just because a packet is missing from a capture.

What Should We Retain About IKE and IPsec Establishment?

Peers follow policy	• IPsec peers may be hosts or security gateways. • The SPD selects traffic for protection; IPsec SAs supply the state ESP uses.
IKE SA and CHILD SA have different jobs	• The authenticated IKE SA protects IKE negotiation and management exchanges. • A CHILD SA contains directional IPsec SAs for selected traffic.
The initial exchanges establish trust and protection	• IKE_SA_INIT agrees on IKE parameters and exchanges DH values and nonces. • IKE_AUTH authenticates the peers and binds the earlier exchange to their identities. • The first CHILD SA is normally established during IKE_AUTH.
Security state is maintained	• CREATE_CHILD_SA can add a CHILD SA, rekey a CHILD SA, or rekey the IKE SA. • IKE SA rekey changes keys without repeating peer authentication.
Diagnosis follows the last confirmed stage	• Compare the capture, installed SA state, selectors, and packet path at both peers. • NAT traversal applies when translation lies on the path; it is not inherent to every deployment.

References and Further Reading

TEXTBOOK READING

William Stallings, Cryptography and Network Security: Principles and Practice

Chapter 20, "IP Security," especially §20.5, "Internet Key Exchange."

Provides a guided explanation of IPsec and IKE before reading the protocol specifications.

PROTOCOL REFERENCES

RFC 7296 — Internet Key Exchange Protocol Version 2 (IKEv2)

Initial exchanges, authentication, CHILD SAs, rekeying, NAT detection, and errors.

RFC 2409 — The Internet Key Exchange (IKE)

Historical IKEv1 specification: Main Mode, Quick Mode, and the two-phase model.

RFC 4301 — Security Architecture for the Internet Protocol

IPsec policy and security associations.

RFC 3948 — UDP Encapsulation of IPsec ESP Packets

ESP over UDP, the Non-ESP Marker, and NAT keepalives.

Lecture capture material: supplied IKEv1 and IKEv2 packet traces examined on Slide 9.